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Introduction

The Town of Hadley requested the Pioneer Valley Planning Commission (PVPC) to conduct a Vulnerable Road User (VRU) Safety study at the intersection of Bay Road (Route 47) and Middle Street (Route 47). The goal of this study is to identify current problems with transportation safety at this intersection with a special focus on multimodal traffic safety. This report details the process of the Vulnerable Road User Safety Assessment (VRUSA) and highlights the key findings of the VRUSA. The following sections provide an overview of the study area, a description of existing conditions around the study area, and a series of recommendations to improve safety for all roadway users.

Figure 1: Aerial View of the Study Area (Google Maps)



Study Area

The intersection of Bay Road and Middle Street is an unsignalized intersection located in the southern area of the Town of Hadley and there is mixed residential and agricultural development in the vicinity of the two intersecting roadways. Hopkins Academy (Town's Middle School and High School) is located in the north of the intersection along Middle Street. Both approaches of Middle Street are controlled by Stop signs while Bay Road has unrestricted traffic flow. A crosswalk is located across the westbound approach of Bay Road.

The westbound approach of Bay Road is a part of Massachusetts Route 47 which continues in the northern direction through this intersection along the southbound approach of Middle Street. Route 47 is classified as Urban Principal Arterial. The eastbound approach of Bay Road is classified as an Urban Minor Arterial, and the northbound Middle Street is a local road. The Hadley Department of Public Works office and yard is located at the southern end of Middle

Street which is a major trip generator for heavy vehicles. There is a single travel lane in each direction of travel along both roads at this intersection. The speed limit along all four approaches of this intersection is 30 mph in the immediate vicinity of the intersection, however, there is a 40-mph speed limit sign located along Bay Road west of its intersection with Middle Street. This sign is visible to westbound traffic along Bay Road further east of the intersection and it appears to increase the speed limit several feet in advance of the actual change in the speed limit farther west of the study area.

Figure 2: Speed Limit Sign Along Westbound Bay Road in the Vicinity of the Intersection

(Speed Limit increases from 30 mph to 40 mph further along the road)



A pole mounted flashing warning beacon signal with a low height Stop Sign is located along the southbound approach of Middle Street. North of the intersection, a flashing red light is displayed for the Middle Street approaches and a flashing yellow light for both approaches of Bay Road. This pole is located in a textured median which appears to have been raised in the past but is almost at the same height as the pavement on Middle Street. This pole is one of the major concerns and safety hazards at this location and has been hit several times in the past by heavy vehicles and farm equipment during single vehicle crashes.

There are two PVT routes that serve Bay Road during the academic year. A bus stop exists on each side of Bay Road on its westbound approach in the vicinity of the intersection.

Figure 3: Pole Mounted Flashing Beacon in the North of the Intersection



Vulnerable Road User Amenities

There are bituminous sidewalks separated by a very wide vegetation buffer with large trees along the eastern and western sides of Middle Street north of the intersection. The sidewalk along the western side does not continue all the way to the intersection of Middle Street with Bay Road. The sidewalk along the eastern side of Middle Street is connected to the sidewalk along the northern side of the westbound approach of Bay Road. A crosswalk along the westbound approach of Bay Road connects the sidewalks along both approaches of Middle Street. A very short section of sidewalk exists along the northbound approach of Middle Street. There are no sidewalks along the eastbound approach of Bay Road.

The intersection does not have any bicycle accommodations. The roads have narrow shoulders and the sidewalks are not wide enough to accommodate bicycles like a shared use path or a side path. This intersection provides access to Skinner State Park trails and trails along the Connecticut river. This location is also part of the network that connects with the newly designated New England Trail¹ which is very popular for mountain biking enthusiasts.

¹ <https://newenglandtrail.org/>

Figure 4: Crosswalk Across Bay Road



Existing Conditions

This section provides a technical evaluation of the transportation components for the intersection. It includes a presentation of the data collected, analysis of traffic operations, and a series of observations and conclusions derived from the analysis.

Average Daily Traffic

The Pioneer Valley Planning Commission (PVPC) collected daily traffic counts and speed data along Bay Road and Middle Street in September 2025. The volumes obtained from the counts have not been adjusted with seasonal adjustment factors for the purpose of analyses in this report. It is possible that some of the results in the warrant analysis or level of service analysis might change if these factors were applied. **Appendix 2** summarizes the traffic volume data separated by direction of travel along each approach. Table 1 represents the Average Daily Traffic separated by direction of travel.

Table 1: Average Daily Traffic

	Northbound	Southbound	Eastbound	Westbound	Total
Bay Road Westbound <i>East of intersection</i>			6,114	5,901	12,015
Bay Road Eastbound <i>West of intersection</i>			5,607	5,497	11,104
Middle Street Southbound <i>North of intersection</i>	737	1,065			1,802
Middle Street Northbound <i>South of intersection</i>	153	151			304

**Values in gray represent volumes entering the intersection*

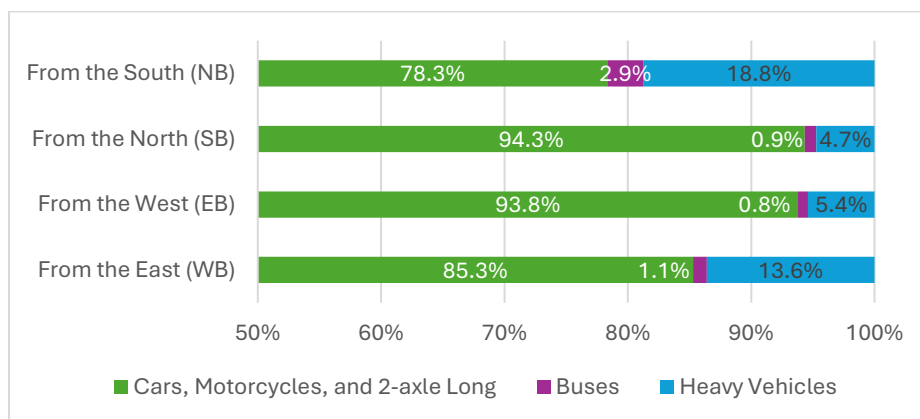
Traffic flow is relatively balanced in both directions of Bay Road. The similar eastbound and westbound traffic volumes along Bay Road indicate higher commuter routes or through traffic going towards Northampton or Amherst/South Hadley.

Vehicle Classification

Vehicles are classified based on the number of axles, number of tires, and the distance between each axle. Two axle - six tire vehicles and vehicles with three or more axles are classified as a “truck” or heavy vehicle. The percentage of heavy vehicle traffic on a roadway is important, as large vehicles have different operating characteristics than normal passenger vehicles. This information is also an important factor in the pavement design of a roadway.

The highest percentage of heavy vehicles has been recorded on the northbound approach of Middle Street. This is directly attributed to the Department of Public Works office and yard located at the end of Middle Street. The southbound and eastbound approaches to the intersection have similar vehicle classification ratios. Data for the westbound approach of Bay Road shows a higher percentage of heavy vehicle traffic than the opposing eastbound direction.

Figure 5: Vehicle Classification by Approach Direction



Vehicle Speed

Appendix 3 depicts the data for the travel speed of vehicles collected by the PVPC in the vicinity of the intersection separated by direction of travel. Table 2 depicts the average and the 85th percentile travel speeds along all four approaches of the intersection. The 85th percentile speed is the speed at which 85 percent of vehicles are travelling at or below. It is one of the key metrics utilized in determining travel speed trends along roadways because of its significance in speed limit regulations.

The travel speed of vehicles in the vicinity of the intersection was separated by direction of travel. The speed limit along both Bay Road and Middle Street is 30 miles per hour. This speed limit reduction occurs approximately 260 feet away from the intersection to the west, and approximately 740 feet away to the east. There is a speed limit reduction to 30 miles per hour on the northern node of Middle Street approximately 450 feet away from the intersection to the north. The flashing beacon assists in alerting drivers about intersecting traffic. Figure 6 depicts the existing Speed limits in the vicinity of the Study Area. This information was gathered from MassDOT GEODOT Roadway Inventory Files and was cross referenced with Speed Limit Ordinance information obtained from MassDOT.

Figure 6: Speed Limit Map in the Vicinity of the Study Area Intersection

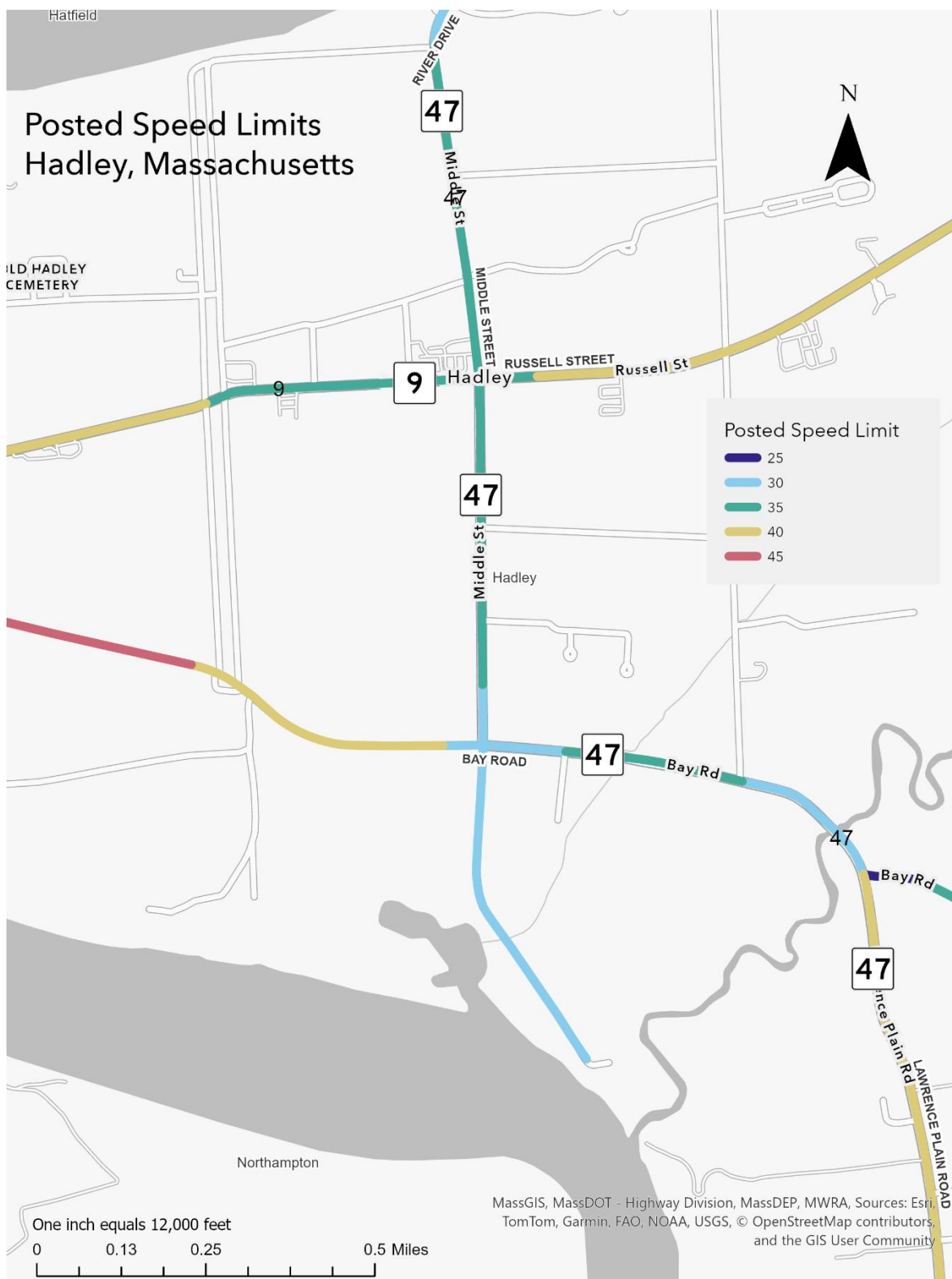


Table 2 represents the average speeds and the 85th percentile speeds along these roads broken down by travel direction. The average speed of vehicles approaching the intersection from both directions along Bay Road is about 10 miles per hour greater than the posted speed limit. 85th percentile speeds are even greater, with some eastbound approaches reaching 44 miles per hour. Speeds on Middle Street tend to be more within range of the speed limit, due in part to the stop signs on these approaches to the intersection.

Table 2: Mean Speeds and 85th Percentile (85%) Speeds in Miles per Hour (mph)

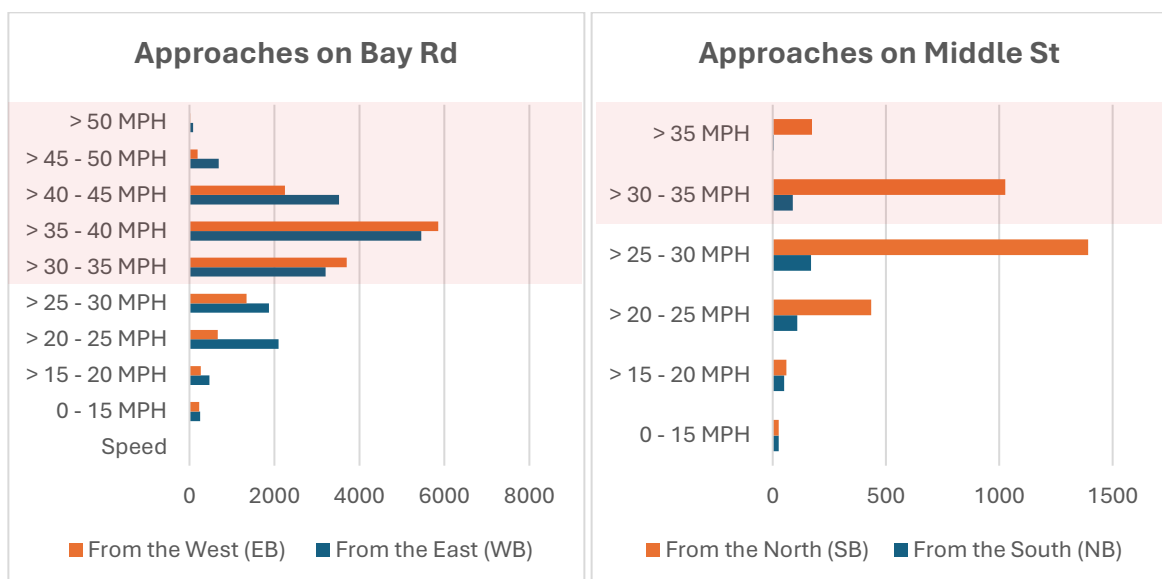
	Posted Speed Limit	Northbound		Southbound		Eastbound		Westbound		Combined	
		Avg. Speed	85% Speed	Avg. Speed	85% Speed	Avg. Speed	85% Speed	Avg. Speed	85% Speed	Avg. Speed	85% Speed
Bay Rd Westbound <i>E/O Middle St</i>	30 mph					33.5 mph	38 mph	34.8 mph	41 mph	34.1 mph	39 mph
Bay Rd Eastbound <i>W/O Middle St</i>	30 mph					35.2 mph	39 mph	37.4 mph	41 mph	36.2 mph	40 mph
Middle St Northbound <i>N/O Bay Rd</i>	30 mph	31.4 mph	34 mph	29.2 mph	32 mph					30.3 mph	33 mph
Middle St Southbound <i>S/O Bay Rd</i>	30 mph	25.6 mph	30 mph	25.6 mph	29 mph					25.6 mph	30 mph

Values in grey represent speeds entering the intersection

Values in red represent average speeds more than 5 mph higher than the Posted Speed Limits

Speed data along Bay Road and Middle Street was further classified into 5-mile-per-hour ranges to identify the percentage of traffic in each range. This information is presented in Figure 7 below. The sections in the colored area of the charts represent the vehicles travelling above speed limit.

Figure 7: Number of vehicles by travel speed along Bay Road and Middle Street



On Bay Road, it is observed that 83 percent of vehicles along the eastbound approach and 73 percent along the westbound approach are traveling above 30 miles per hour. 17 percent and 24 percent of eastbound and westbound traffic respectively are observed to be exceeding 40 miles per hour. On Middle Street, speeding is observed but is less prevalent than on Bay Road; 21 percent of northbound traffic and 39 percent of southbound traffic is seen to be exceeding 30 miles per hour on Middle Street.

Public Transit

There are two fixed bus routes operated by the Pioneer Valley Transportation Authority (PVTA) that serve Bay Road. One route is the 39, which provides service between Smith College in Downtown Northampton and Hampshire College in Amherst via Route 9 and Bay Road, with some trips deviating to Hampshire Mall on Route 9 in Hadley. The other route is the 39E, providing trips between Smith College and Mt. Holyoke College via Route 9, Bay Road, and Route 47. Both routes only operate during the academic year. There is a pair of bus stops on Bay Road just east of the intersection, one in each direction of travel. Table 3 shows the ridership totals (totals for entire time period) for February through May of 2025, including boards, alights (departing the bus), trips operated, and days operated for each of the two bus stops.

Table 3: PVTA Ridership Data at Bay Road Bus stops, Feb-May 2025

Eastbound stop	February	March	April	May
Boards	2	3	5	2
Alights	3	2	0	0
Trips Operated	219	173	237	88
Days Operated	20	16	22	8
Westbound stop	February	March	April	May
Boards	0	0	3	0
Alights	8	3	9	0
Trips Operated	221	174	237	88
Days Operated	20	16	22	8

Source: PVTA

Vulnerable Road User Counts

The Federal Highway Administration defines a vulnerable road user (VRU) as a non-motorist such as a “pedestrian, bicyclist, other cyclist, and person on personal conveyance. A vulnerable

road user may include people walking, biking, or rolling.² This may include a worker in a construction work zone. The PVPC collected data for pedestrians, bicyclists, and other vulnerable road users at this intersection by utilizing a low-resolution video camera to capture video footage along the study area for a period of four days. This exercise resulted in three full days of data with one partial day – a total of 60 hours of video footage.

A total of 124 VRUs were observed throughout the study area during this period. 80 pedestrians, 43 bicyclists, and one scooter user were observed. Figure 8 depicts the hourly distribution of ADT (Average Daily Traffic of VRUs) and Figure 9 depicts their location and movement patterns for the entire duration of 3 days of the data collection period.

Figure 8: Vulnerable Road User Count (Average Daily Count) During Hours of Daylight

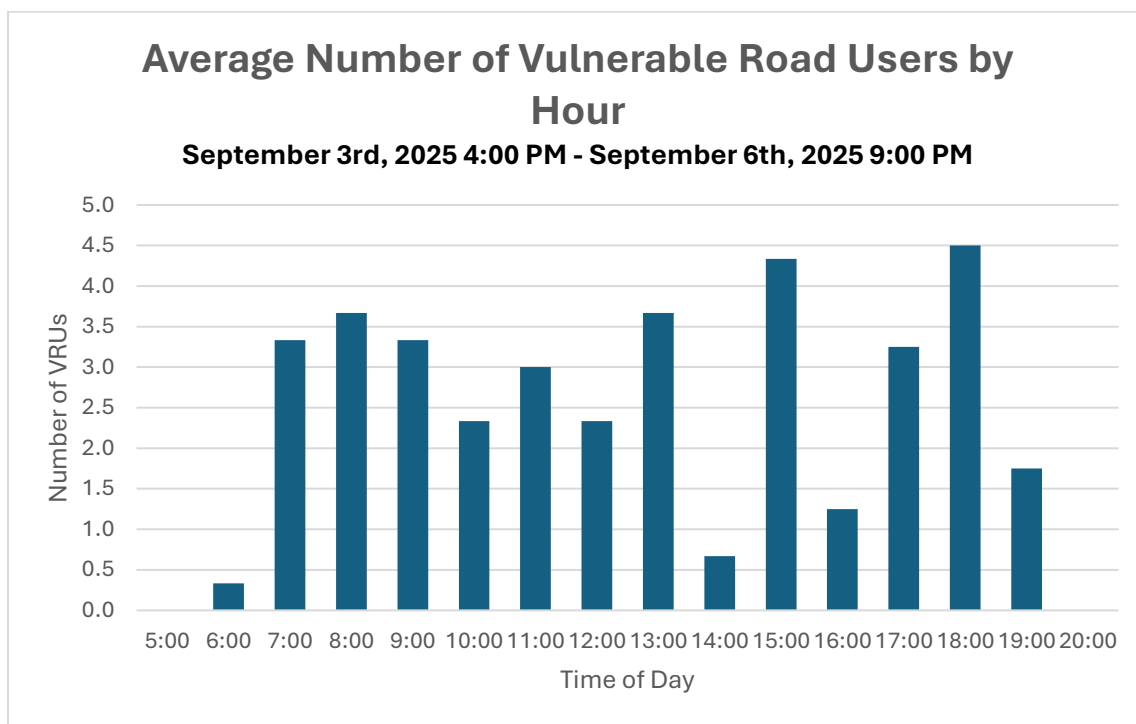
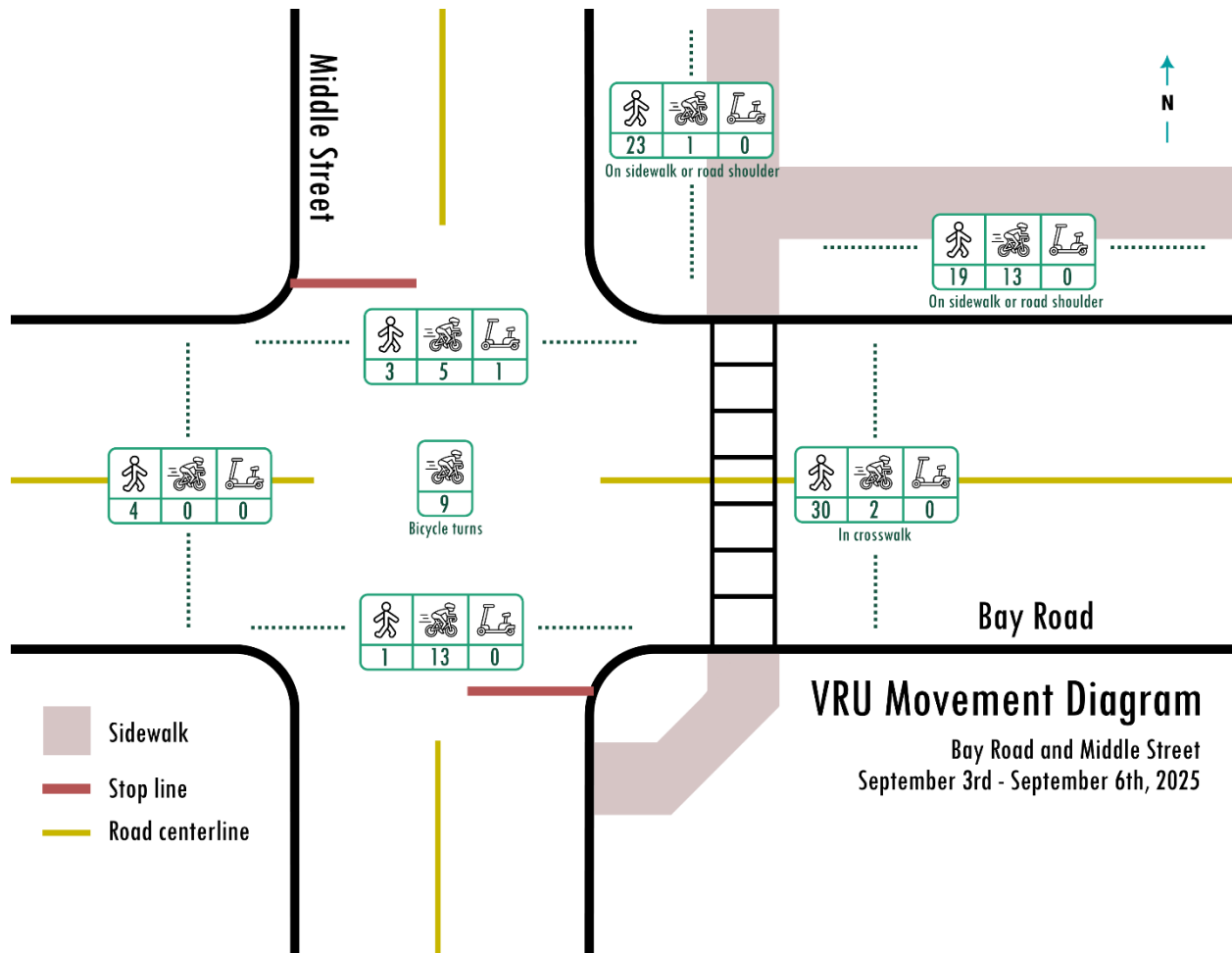


Figure 9 indicates a significant number of pedestrians and bicyclists cross the pavement not utilizing the crosswalk. This is unsafe behavior, especially along the eastbound approach of Bay Road which does not operate under STOP control. The figure also illustrates a high number of bicyclists (almost 35% of the total VRUs) along this location even though both the corridors at this intersection in the study area lack bike lanes and have narrow shoulders.

² https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-10/VRU%20Safety%20Assessment%20Guidance%20FINAL_508.pdf

Figure 9: VRU (Total Counts for 3 Days) Location and Movement



Peak Hour Turning Movements

Turning Movement Counts (TMCs) were conducted for the intersection during the morning and afternoon peak hours. The weekday peak commuter period occurs during the morning hours of 7:00 AM to 9:00 AM and the afternoon hours of 2:00 PM and 6:00 PM. The TMCs were conducted to identify the peak four consecutive 15-minute periods of traffic through the intersection. These consecutive peak periods constitute a location's Peak Hour Volume. The peak hour of traffic volume represents the most critical period for operations and will be the focus for some of the analysis conducted in this study.

Included in the TMC study is the identification of heavy vehicles on the roadway. Heavy vehicles include trucks, recreational vehicles and buses. The percentage of heavy vehicles in the traffic flow is an important component in calculating the serviceability of a corridor or intersection. Trucks occupy more roadway space than passenger cars and can impact traffic flow. Their larger size and slower acceleration, deceleration, and maneuverability make them less responsive than

smaller vehicles. This impact is factored in when determining the level of service for the intersection.

Figures 10 and 11 depict the results of the TMCs during each peak hour period. Figure 10 represents the peak morning commuting hour between 8:00 AM and 9:00 AM, and Figure 11 represents the peak afternoon commuting hour between 2:45 PM and 3:45 PM.

Bay Road carries roughly 7–8 times more traffic than Middle Street. There is higher volume in the afternoon by approximately (6% overall) along three of the approaches except the Middle Street southbound approach. The morning peak has a higher number of right turns from Bay Road onto Middle Street going northbound along Route 47, which coincides with school pick-up or commuter traffic patterns. Bay Road westbound through movements decrease in the afternoon, while eastbound throughs increase which also indicates commuter traffic in the afternoon.

Figure 10: Peak Morning Turning Movement Counts 8:00 am to 9:00 am

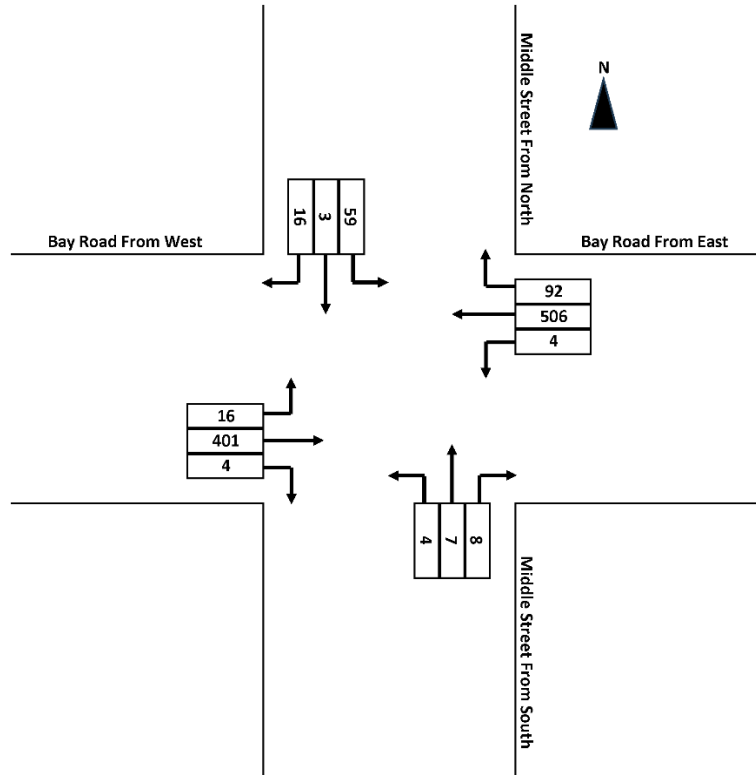
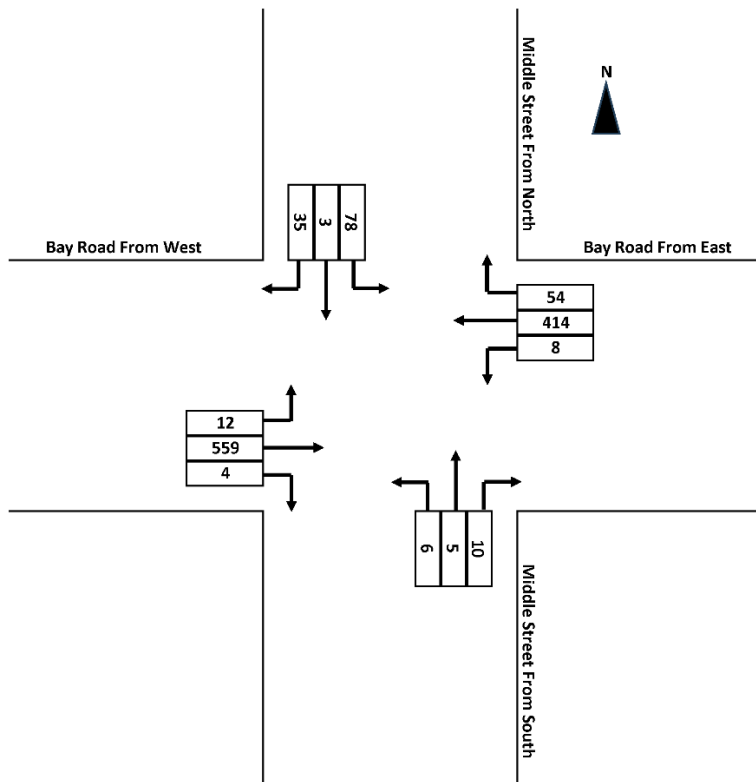


Figure 11: Peak Afternoon Turning Movement Counts 2:45 pm to 3:45 pm



Congestion and Level of Service Analysis

The intersection was examined regarding capacity and delay characteristics to determine the existing Level of Service (LOS). LOS is an indicator of the operating conditions which occur on a roadway under different volumes of traffic and is defined in the Highway Capacity Manual by six levels, A' through 'F'. Several operational factors can influence the LOS including geometry, travel speeds, delay, and the number of pedestrians. Depending on the time of day and year, a roadway may operate at varying levels. Level of Service 'A' represents the best operating conditions and is an indicator of ideal travel conditions with vehicles operating at or above posted speed limits with little or no delays. Conversely, LOS 'F', or failure, generally indicates forced flow conditions illustrated by long delays and vehicle queues. Level of service 'B' is observed when the traffic flows relatively smoothly without significant slowdowns or increased delays to travel time. Level of Service 'C' indicates a condition of stable flow and is generally considered satisfactory in rural areas. Under LOS 'D' conditions, delays are considerably longer than under LOS 'C' but are considered acceptable in urban areas. At LOS 'E' the roadway begins to operate at unstable flow conditions as the facility is operating at or near its capacity. Table 4 depicts the delay and LOS designations along unsignalized intersections as per the Highway Capacity Manual. Table 5 depicts the results of PVPC's LOS Analysis.

Table 4: LOS Designations

Level of Service	Expected Delay to Minor Street	Average Control Delay (s/vehicle)
A	Little or no delay	0.0 to 10.0
B	Short Traffic Delays	>10.0 to 15.0
C	Average Traffic Delays	>15.0 to 25.0
D	Long Traffic Delays	>25.0 to 35.0
E	Very Long Delays	>35.0 to 50.0
F	Extreme Delays	>50.0

Overall, the intersection Level of Service is A for both peak hours because Bay Road traffic, which constitutes a vast majority of the volume, does not experience delay; however, Middle Steet experiences higher congestion especially for the southbound approach in the morning and the northbound approach in the afternoon. This finding is consistent with some queueing observed along the Middle Street southbound approach during the afternoon peak hour at the time of field visit during Hopkins Academy school end time. Based on the LOS data, it can be inferred that queueing likely occurs on Middle Street regularly, particularly during school pick-up and drop-off periods.

Table 5: Existing Level of Service

Approach	AM Peak Hour		PM Peak Hour	
	Delay	Level of Service	Delay	Level of Service
Bay Road Westbound	0.1	A	0.3	A
Bay Road Eastbound	0.5	A	0.3	A
Middle Street Southbound	34.0	D	21.4	C
Middle Street Northbound	19.4	C	44.7	E

In the following sections PVPC will analyze both a Four-Way Stop Warrant and Traffic Signal Warrant for the intersection. This analysis will compare standard LOS service scenarios under existing traffic volume levels for both a Four-Way Stop and Traffic Signal operations.

Warrants

Multi-Way Stop or Four-Way Stop Warrants

The Manual on Uniform Traffic Control Devices (MUTCD)³ Section 2B.12-17 contains warrants for multiway stop control installations. The following criteria are considered for determining use of a multiway stop control:

The decision to establish all-way stop control at an unsignalized intersection should be based on an engineering study. The engineering study for all-way stop control should include an analysis of factors related to the existing operation and safety at the intersection, the potential to improve these conditions, and the applicable factors contained in the following all-way stop control warrants:

- A. Crash Experience – 5 or more crashes in 12 -month period or 6 or more crashes in 36-month period of a type correctable by the installation of All-Way Stop signs.
- B. Sight Distance - All-way stop control may be installed at an intersection where an engineering study indicates that adequate sight distance from the minor-road does not exist.
- C. Transition to a Traffic Signal Control or Transition to Yield Control at a Circular Intersection - All-way stop control may be installed at locations where all-way stop control is an interim measure prior to the installation of a traffic signal or a roundabout.

³ https://mutcd.fhwa.dot.gov/kno_11th_Edition.htm

D. 8-Hour Volume – A minimum of 300 vehicles, bicycles and pedestrians per hour for 8 hours from the major street approach and a minimum of 200 vehicles, bicycles and pedestrians per hour for 8 hours from the minor street approach for the same 8 hours.

E. Other Factors as determined by an engineering study.

MassDOT has still not adopted the new MUTCD guidelines; however, the volume and the number of annual crashes under MassDOT's current guidelines for a Four-Way Stop Warrant are similar to the above-mentioned thresholds. The study area intersection does not meet the crash or the volume requirements for a Four-way Stop control. Under current conditions a Four-way Stop does not resolve the issues at this location, therefore alternate improvement measures should be considered.

Traffic Signal Warrants

Under standard guidelines, a traffic signal is a step up from a Multi-Way Stop if the latter does not improve or alleviate traffic issues along an intersection. However, under certain unique circumstances, especially at locations where traffic issues are concentrated during specific times of day, a traffic signal is considered as a potential solution. Even though an intersection may not have traffic volumes high enough to warrant a Multi-Way Stop, it might still reach the thresholds for peak hour volume requirements of a traffic signal. That being said, installing a traffic signal at an intersection should only be considered based on a thorough engineering evaluation of a location. The MUTCD states that a traffic signal should be installed only after all other measures and improvements to improve traffic conditions at an intersection have failed to solve the problems of congestion and safety.

The Manual on Uniform Traffic Control Devices (MUTCD)⁴ identifies nine different warrants to evaluate if an intersection meets the minimum requirements for signalization. One or more warrants must be satisfied to justify the installation of a traffic signal; however, engineering judgment ultimately dictates if an intersection warrants the installation of a traffic signal. The installation of a traffic signal must also improve the safety and operation of the location under study. Of the nine total warrants for the installation of a traffic signal, Warrant 1 – Eight Hour Vehicular Volume is generally considered the most important as it requires minimum volumes to be met on both the major and minor streets for at least eight hours. Warrant 2 – Four Hour Vehicular Volume and Warrant 3 – Peak Hour Volume also require minimum volumes to be met but over shorter timeframes. Warrant 4 – Pedestrian Volume Warrant is specifically designed to address the needs of pedestrians at a given intersection by establishing guidelines for

⁴ https://mutcd.fhwa.dot.gov/kno_11th_Edition.htm

installation of a traffic signal if the volume of pedestrians crossing is above the thresholds provided in the warrant. Warrant 7 – Crash Experience requires 80% of the volume requirements of Warrant 1 to be satisfied along with a history of minimum crash thresholds defined by crash type and severity. In addition to all that, Warrant 7 also requires a demonstration that less restrictive remedies such as improved signage and pavement markings be tried and have failed to reduce crashes before a signal can be installed.

Table 6 represents the results of the Signal Warrant Analysis (SWA) conducted utilizing Highway Capacity Software (HCS) 2023. The intersection meets the volume requirements for Warrants 2 and 3. Both these Warrants are intended to evaluate possible improvements along intersections where the volume of cross-traffic or minor street traffic is the primary reason to consider installing a traffic signal. In this scenario the congestion and queueing along Middle Street during limited times of the day is triggering the Warrants.

Simply meeting these numbers does not automatically require a signal to be installed. An engineering study and professional judgment are still required to determine if a traffic signal is the best solution for the intersection.

Table 6: Results of Traffic Signal Warrant Analysis

	Description	Result
Warrant 1	Eight – Hour Vehicular Volume	Not Satisfied
Warrant 2	Four- Hour Volume	Satisfied
Warrant 3	Peak Hour Volume	Satisfied
Warrant 4	Pedestrian Volume	Not Satisfied
Warrant 5	School Crossing	N/A
Warrant 6	Coordinated Signal System	N/A
Warrant 7	Crash Experience	Not Satisfied
Warrant 8	Road Network	N/A
Warrant 9	Intersection Near a Grade Crossing	N/A

N/A = Not Applicable

Evaluating Congestion and LOS under Alternate Improvement Scenarios

The PVPC analyzed the level of service under a Multi-Way Stop Control and a Traffic Signal Control along this intersection utilizing peak hour volumes collected at the time of this study. The results are represented in the Tables below.

Table 7: LOS Analysis under Four-Way Stop Control

Approach	AM Peak Hour		PM Peak Hour	
	Delay	Level of Service	Delay	Level of Service
Bay Road Westbound	17.0	C	36.8	E
Bay Road Eastbound	31.1	D	22.7	C
Middle Street Southbound	9.9	A	10.3	A
Middle Street Northbound	10.8	B	11.8	B

As can be seen in the above Table, a Four-Way Stop is not a potential solution for improving congestion along this intersection as it shifts the delay from minor street approaches to Bay Road. Bay Road has significantly higher traffic than Middle Street and increasing delay along Bay Road does not improve but rather worsens the traffic delay at this location.

Table 8: LOS Analysis under an Actuated 4-Phase Traffic Signal

Approach	AM Peak Hour		PM Peak Hour	
	Delay	Level of Service	Delay	Level of Service
Bay Road Westbound	16.2	B	30.4	C
Bay Road Eastbound	32.9	C	19.0	B
Middle Street Southbound	6.8	A	6.6	A
Middle Street Northbound	7.9	A	7.6	A

The LOS designations based on approach or movement delay in seconds are different for a signalized intersection compared to unsignalized intersections. Slightly higher approach delays are allowable to optimize over all cycle length for an intersection. An actuated traffic signal could optimize operations at this intersection without creating an overly congested scenario along major street or minor street approaches. However, it is once more worthy to note that a traffic signal should only be considered after all other measures to improve conditions have been tried and have failed to alleviate the problems and issues.

Safety

To analyze the issues related to traffic safety and crashes, the PVPC evaluated the historic crash data for past 10 years obtained from MassDOT's IMPACT database. This data clarified historic trends. On the other hand, to understand the factors influencing individual crashes and to obtain crash patterns, the PVPC obtained crash report copies from the Hadley Police Department for the calendar years of 2021 to 2025 (through April).

Historic Crash Data

A total of 13 crashes occurred at this intersection during the 10-year period between 2015 to 2024 (Table 9). A vast majority of those crashes occurred during clear weather and dry road conditions during the afternoon peak travel times. Only 3 crashes occurred when the road was wet, and it was cloudy or raining. The charts in Figures 12 and 13 depict the crash severity and manner of collisions trends.

Table 9: Annual Crashes in Last 10 years

2015	1
2016	3
2017	1
2019	1
2021	1
2022	4
2023	1
2024	1
Total	13

Figure 12: Manner of Collision

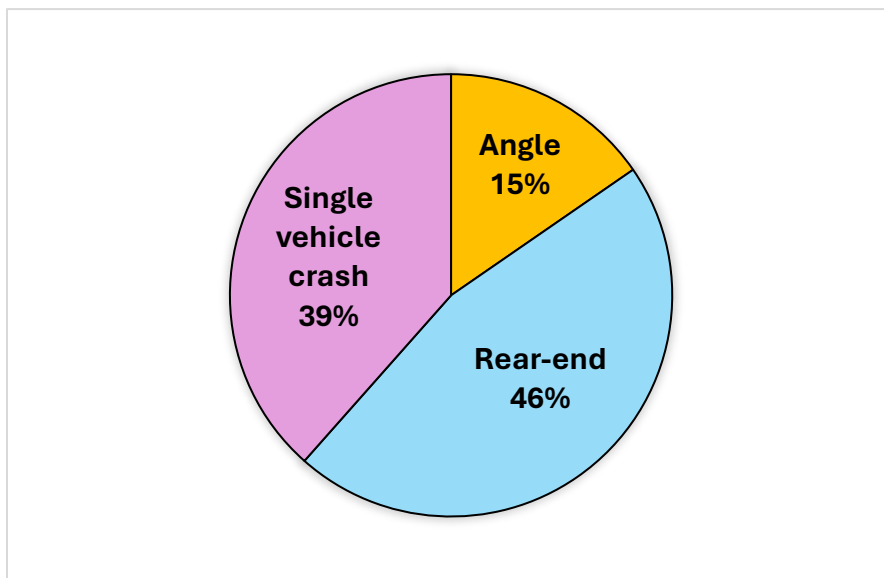
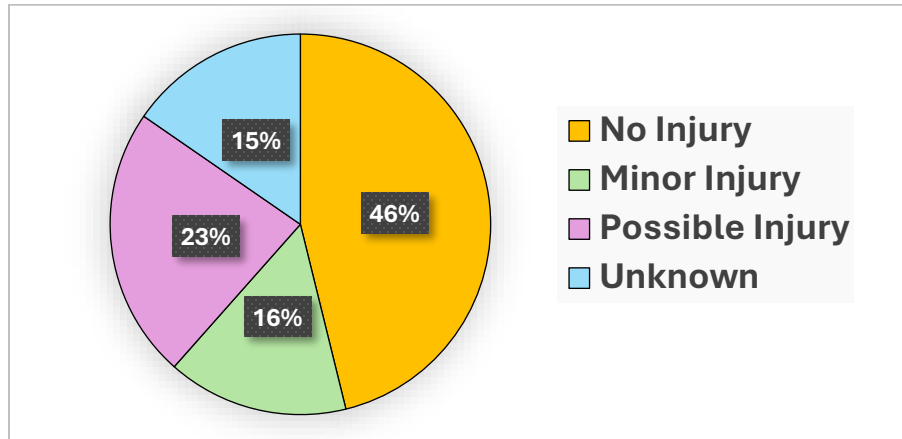


Figure 13: Crash Severity



A high number of rear end collisions occur when speeding is an issue and vehicles are not stopping in time to avoid colliding with vehicles ahead of them slowing down or stopping to yield to other traffic. The high percentage of single vehicle crashes at this intersection is an unusual pattern for such location. As mentioned earlier in the Study Area introduction, these crashes confirm the collisions with the pole mounted flashing beacon or the protective bollards in the vicinity of the median on Middle Street.

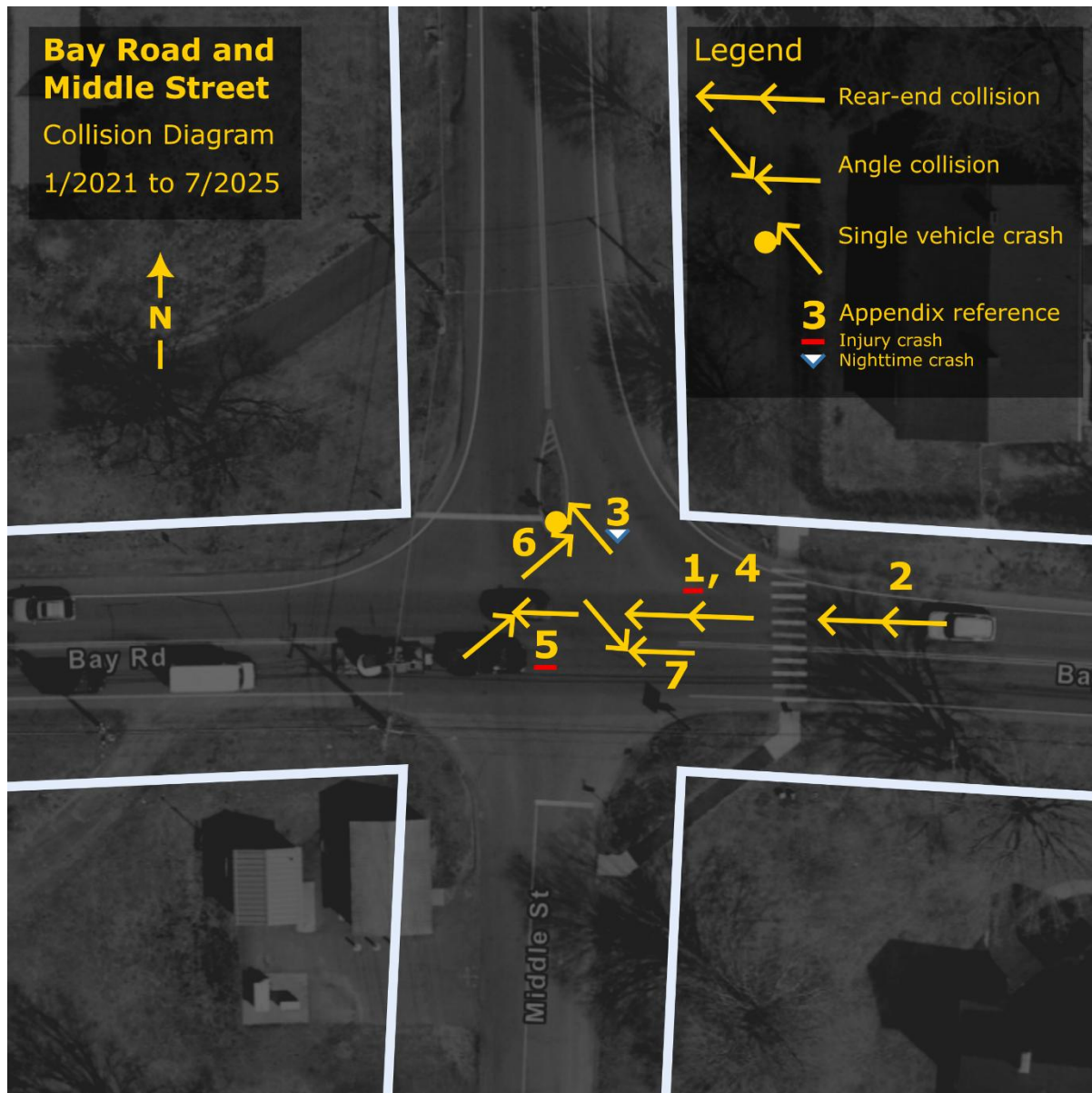
Collision Diagram

PVPC obtained the crash reports from the Hadley Police Department to develop a collision diagram, The objective was to analyze the collision patterns and determine the factors that contribute to an increased number of collisions at this intersection.

In this period of time, there were a total of seven crashes that had occurred at this intersection. Three of these crashes were rear-end collisions; another two were angle collisions, and the remaining two were single-vehicle collisions. This trend is evident from the IMPACT crash data analysis as well. Each crash has been depicted graphically on the collision diagram based on the details provided in the crash report for each incident. Figure 14 depicts these crashes.

It can be seen from the collision diagram that the westbound approach to the intersection has the highest percentage of incidents out of all four approaches. Two of these westbound approach collisions were rear-end collisions resulting from vehicles slowing or stopping while attempting to turn; another was the result of a vehicle stopping at the marked crosswalk to give way to pedestrians waiting to cross Bay Road. The final westbound collision occurred when a vehicle traveling westbound crossed the divider line to pass a turning vehicle in front, leading to an angle collision with another vehicle turning left onto Bay Road from the southbound approach of Middle Street.

Figure 14: Collision Diagram



Another prominent finding of the collision diagram is the two single-vehicle collisions at the flashing beacon on the island of Middle Road. Two of these collisions occurred, with one vehicle turning right onto Middle Street whose driver was and the other turning left, both from Bay Road. The vehicle turning left was a large trailer truck. The driver operating the right turning vehicle that struck the beacon was impaired. No incidents involving vulnerable road users were reported during the study period. MassDOT informed the PVPC of an additional crash with the flashing beacon that occurred during this time period which was not listed in the local crash data obtained from Hadley Police Department.

Vulnerable Road User Safety Assessment (VRUSA)

To understand the VRU safety issues and to obtain input from stakeholders, the PVPC organized a Vulnerable Road User Safety Assessment (VRUSA) in collaboration with the Town of Hadley, MassDOT, and MassBike. A team of multidisciplinary stakeholders visited the study area to assess the conditions, determine the issues, and discuss potential recommendations to improve multi modal traffic conditions in the area

Details

The VRUSA was conducted on Thursday August 14th, 2025, between 9:00 am and noon, at the Hadley Police Department. It was attended by 10 people. The names of the attendees and the organizations they represent are summarized in Table 10.

Table 10: VRUSA Attendees

No.	Name	Organization	Email
1	Scott McCarthy	Hadley DPW	mcCarthyS@hadleyma.gov
2	Mitch Kuc	Hadley Police	kucm@hadleyma.gov
3	Jane Nevinsmith	Hadley Selectboard	nevinsmith2@charter.net
4	Alexis Hosea-Abbott	Mass Bike	alexis@massbike.org
5	Sarah Cannamela	MassDOT D2	sarah.a.cannamela@dot.state.ma.us
6	Tom Ruta	MassDOT D2	thomas.ruta@dot.state.ma.us
7	Jeff McCollough	PVPC	jmccollough@pvpc.org
8	Cole Hartford	PVPC	chartford@umass.edu
9	Geoffrey Klafeta	PVPC	gklafeta@pvpc.org
10	Khyati Parmar	PVPC	kparmar@pvpc.org

The Agenda for the meeting is attached as Appendix 4 and the Sign-in sheet is included in Appendix 5. The team assembled at the Police Department and visited the site following a brief discussion of issues. They walked along the study area and surveyed all four approaches from a VRU perspective. They returned to the Police Department after the site visit to discuss improvement measures pertaining to observed problems.

Summary of Issues, Comments, and Potential Improvements

Table 11 summarizes the discussion during the VRUSA in terms of issues and recommendations to resolve those concerns. During the discussions, the team brainstormed all possible ideas and

alternatives which are summarized here; however, the execution or implementation of the best suited improvement measures is subject to a thorough and detailed engineering study and assessment of short- and long-term implications of the measure.

Table 11: VRUSA Notes

Issues	Comments	Discussion
Speeding	• People drive at a higher speed along Bay Road, which is major street at this intersection with higher traffic. • The flashing beacon light post on the Middle Street Island is not very effective at traffic calming. • Heavy vehicles travel fast along this intersection • There is more traffic during school year and younger drivers tend to speed.	• Considerations were made that the Town could consider a 25 MPH townwide speed limit. • It was discussed to consider if thickly settled or business district speed limit ordinance could be applied at this intersection to reduce speed limit from 30 mph but it's unlikely that the conditions are met for any of these designations due to lower density. • Four-way stop, and Traffic Signal warrants were discussed. • An overhead / hanging flashing beacon should be considered because the current flashing yellow light is out of the way and is easily ignored. • Horizontal / lateral rumble strips are an option. Shallower to avoid damage due to icing in winter and to not disrupt snowplows. • Narrowing down the traffic lanes could help to reduce speeding. • Speed humps or a raised crosswalk are also options for speed calming to be considered but both shouldn't be too tall so they're not a safety hazard. • Discussion was held regarding MassDOT's the pilot program for horizontal rumble strips on Damon Road and why MassDOT does not recommend them.

Issues	Comments	Discussion
Visibility	• Visibility is bad driving northbound from middle street when you stop on the line, cars must creep towards the intersection to pull out. • This approach provides access to DPW and DPW service vehicles and heavy trucks and equipment utilize this approach. • The beacon on the island on Middle Street is difficult to see from the westbound Bay Road approach. • The stop lines are uneven on Middle Street northbound vs southbound and do not extend along entire width of the approach. • The current crosswalk is not visible enough to drivers on Bay Road.	• The stop line along northbound approach could be moved and extended throughout the travel lane. This is not an issue but this suggestion was made at VRUSA • The beacon could be moved to overhead mast arm or span wire. • The crosswalk could be raised or textured like examples in Northampton and Easthampton. • Raised intersection concept was discussed. • Warning signs could be doubled. • RRFB with overhead pedestrian light was recommended for the crosswalk.
Crosswalk	• The current crosswalk is difficult for drivers to see coming from the eastbound approach as they are driving slightly up a hill. • Drivers oftentimes do not stop at the crosswalk, the Hadley police tested by waiting for the crosswalk themselves and many drivers didn't stop. • The advance crosswalk warning signs are very close to the actual crosswalk. • Kids utilize this crosswalk and intersection during peak hours. • There are children walking, biking, and skateboarding to school along this crosswalk.	• The current crosswalk needs to be improved, either through a raised design, grooves, adding more signage, or just making the markings on the pavement more aggressive. • The crosswalk should be made accessible for the visually impaired or those needing a wheelchair. • A HAWK Signal across Bay Road is an option to make pedestrians in the crosswalk more visible. • Widening existing crosswalk was discussed. • More visible crosswalk markings guidelines in MUTCD were discussed. • Examples to discuss and follow for crosswalks were brought up like the ones around Williston in Easthampton, and Pomeroy Terrace and Jackson Street in Northampton by the school.
Intersection Geometry and Middle Street Island	• The location of the Flashing Beacon is not ideal for visibility along major street / Bay Road. • The island has almost become a textured pavement due to construction of pavement around it. • The pole which supports the beacon and the stop sign has been hit multiple times. • The Bollards on the island on Middle Street need to be repaired often because large trucks and farm equipment come through the intersection and hit them.	• The Island on Middle Street should either be "formalized" with an additional curb so that it stands out or it should be removed entirely so it doesn't need to be upkept when trucks moving through the intersection hit the bollards or flashing light. • A roundabout could be considered for the intersection, but the Town would likely oppose one as the Amherst section of Bay Road already has two roundabouts. • Tightening radii forces vehicles to slow down while turning. MassDOT approved RRFB grant but MUTCD does not advise RRFB at 4-way intersection.

Issues	Comments	Discussion
	- Recently Hadley police witnessed a near-miss at this intersection. This is a common occurrence as near misses are frequently observed at the intersection. - The island on Middle Street is not very helpful because it is off to the side of the intersection.	- A tear drop island in the middle of the intersection was discussed. - Curb extensions / bump outs for the island on Middle Street were discussed. - A raised island with proper curbing will provide a buffer to the pole and bollards and define the turning radii for heavy vehicles. - Changing the turning radii but altering the intersection geometry was discussed - Guidelines for marking turning lane was discussed however those are mostly utilized when there are multiple lanes of turning traffic.
Signage	- The 40 MPH sign facing the westbound approach is very close to the intersection. It is visible to vehicles along westbound approach well in advance of the intersection and this might encourage vehicles to speed up while passing through the intersection. - Sometimes people ignore the stop signs on Middle Street.	- The 40 MPH sign facing the westbound approach should be moved away from the intersection, so cars are not incentivized to speed. - It was discussed that the DPW had plans to install speed feedback signs along Bay Road. The town should proceed with the plan. - Intersection ahead pavement markings and supplemental crosswalk ahead signs should be included at the intersection. - Flashing stop signs could make the stop signs on Middle Street more visible. - One study found that adding flashing LEDs to a stop sign reduced blow-throughs by 52.9% and roll-throughs by 28.9%.
Vulnerable Road Users' Amenities	- The current bus stops nearby are not well developed with no benches or other good places to sit. - The bus stop on the southern side of Bay Road does not have a sidewalk meeting up to it. - The 'Bikeability' rating was described as medium to low, but bikes are frequently observed. - This intersection is an important node along the network of corridors that provide access to New England Trail which is very popular for Mountain Bikers and Bikers alike. Therefore, bike infrastructure is vital in long term. - Schoolchildren utilize bikes and scooters to get to the school. - Tree roots often tear up the nearby sidewalk causing maintenance issues.	- The nearby bus stops should have a bench, be that formalized or unformalized. Coordinate with the PVTa and make changes based on demand. - Marked bike lanes would increase safety for bikers at the intersection. - A pedestrian curb ramp along all legs of the intersection will boost accessibility at the intersection. - A new shared use path at the intersection would make it more accessible to pedestrians. - A separated side path is important for pedestrians and the Town along this location.

Recommendations

The following recommendations are presented for the Town of Hadley to improve VRU safety and overall transportation conditions in the vicinity of this intersection.

Existing Infrastructure

The Town of Hadley could utilize the following short-term measures to make changes to the existing infrastructure and improve traffic operations at this location.

Signage

STOP Sign: Some of the vehicles along the Middle Street southbound approach do not stop at the stop sign. It is recommended that the Town considers installing LED-enhanced STOP signs or STOP signs with flashing borders. The MUTCD 11th Edition (2023) provides more guidance on these Enhanced Stop signs in Part 2A General, Section 2A.07, and Section 2B.05⁵

Crosswalk Ahead / Pedestrian Crossing Advance Warning Signs: The current Crosswalk Ahead Warning Signs along Bay Road for eastbound and westbound approaches are located at an approximately 130 ft distance from the existing crosswalk (Figure 15). This distance is not adequate for vehicles travelling above 30 mph.

Figure 15: Distance of Crosswalk Ahead Advance Warning Signs from the Crosswalk



⁵https://mutcd.fhwa.dot.gov/pdfs/11th_Edition/Chapter2a.pdf

Section 2C-04 and Table 2C-3 (Table 12), "Advance Placement Distance for Warning Signs," within Chapter 2C of MUTCD 11th Edition 2023 provides recommended minimum sign placement distances for various speeds under two different conditions: A and B. Condition A is for complex situations where there are multiple travel lanes while B is for less complex situations where a driver's only required action is to slow down.

Table 12: MUTCD 11th Editions 2023, Table 2C-3

Table 2C-3. Guidelines for Advance Placement of Warning Signs

Posted or 85th-Percentile Speed	Advance Placement Distance ¹									
	Condition A: Speed reduction and lane changing in heavy traffic ²	Condition B: Deceleration to the listed advisory speed (mph) for the condition								
		0 ³	10 ⁴	20 ⁴	30 ⁴	40 ⁴	50 ⁴	60 ⁴	70 ⁴	80 ⁴
20 mph	225 ft	115 ft	N/A ⁵	—	—	—	—	—	—	—
25 mph	325 ft	155 ft	N/A ⁵	N/A ⁵	—	—	—	—	—	—
30 mph	460 ft	200 ft	N/A ⁵	N/A ⁵	—	—	—	—	—	—
35 mph	565 ft	250 ft	N/A ⁵	N/A ⁵	N/A ⁵	—	—	—	—	—
40 mph	670 ft	305 ft	100 ft ⁶	100 ft ⁶	N/A ⁵	—	—	—	—	—
45 mph	775 ft	360 ft	125 ft	100 ft ⁶	100 ft ⁶	N/A ⁵	—	—	—	—
50 mph	885 ft	425 ft	200 ft	175 ft	125 ft	100 ft ⁶	—	—	—	—
55 mph	990 ft	495 ft	275 ft	225 ft	200 ft	125 ft	N/A ⁵	—	—	—
60 mph	1,100 ft	570 ft	350 ft	325 ft	275 ft	200 ft	100 ft ⁶	—	—	—
65 mph	1,200 ft	645 ft	450 ft	400 ft	350 ft	275 ft	200 ft	100 ft ⁶	—	—
70 mph	1,250 ft	730 ft	525 ft	500 ft	450 ft	375 ft	275 ft	150 ft	—	—
75 mph	1,350 ft	820 ft	625 ft	600 ft	550 ft	475 ft	375 ft	250 ft	100 ft ⁶	—
80 mph	1,475 ft	910 ft	725 ft	700 ft	625 ft	550 ft	450 ft	350 ft	200 ft	—
85 mph	1,600 ft	1,010 ft	825 ft	800 ft	750 ft	675 ft	575 ft	450 ft	300 ft	150 ft

¹ The distances are adjusted for a sign legibility distance of 180 feet for Condition A. The distances for Condition B (with the exception of the potential stop condition) have been adjusted for a sign legibility distance of 250 feet, which is appropriate for an alignment warning symbol sign. For Conditions A and B, warning signs with less than 6-inch legend or more than four words, a minimum of 100 feet should be added to the advance placement distance to provide adequate legibility of the warning sign.

² Typical conditions are locations where the road user must use extra time to adjust speed and change lanes in heavy traffic because of a complex driving situation. Typical signs are Merge and Right Lane Ends. The distances are determined by providing the driver a PRT of 14.0 to 14.5 seconds for vehicle maneuvers (2018 AASHTO Policy, Table 3-3, Decision Sight Distance, Avoidance Maneuver E) and adjusted for a legibility distance of 180 feet for the appropriate sign.

³ Typical condition is the warning of a potential stop situation. Typical signs are Stop Ahead, Yield Ahead, Signal Ahead, and Intersection Warning signs. The distances are based on the 2018 AASHTO Policy, Table 3-1, Stopping Sight Distance, providing a PRT of 2.5 seconds, a deceleration rate of 11.2 feet/second².

⁴ Typical conditions are locations where the road user must decrease speed to maneuver through the warned condition. Typical signs are Turn, Curve, Reverse Turn, or Reverse Curve. The distance is determined by providing a 2.5 second PRT, a vehicle deceleration rate of 10 feet/second², and adjusted for a sign legibility distance of 250 feet.

⁵ No suggested distances are provided for these speeds, as the placement location is dependent on site conditions and other signing. An alignment warning sign may be placed anywhere from the point of curvature up to 100 feet in advance of the curve. However, the alignment warning sign should be installed in advance of the curve and at least 100 feet from any other signs.

⁶ The minimum advance placement distance is listed as 100 feet to provide adequate spacing between signs.

Note: Warning signs that advise road users about conditions that are not related to a specific location, such as Deer Crossing or SOFT SHOULDER, can be installed in an appropriate location, based on engineering judgment.

This intersection has a single travel lane along each approach and the 85th Percentile speeds along Bay Road are approximately 40 mph for both directions. Therefore, based on Condition B: Deceleration to the listed advisory speed in the above table, it is recommended that

supplemental Crosswalk Ahead Warning Signs be installed at a distance of minimum 305 ft or more from the existing crosswalk.

Existing 40 mph Speed Limit Sign: This Speed Limit Sign which is intended for westbound traffic along Bay Road is located less than 75 feet west of the intersection. This sign is most likely placed in advance of the actual speed limit change along the corridor and potentially encourages drivers to increase their speed while travelling through this intersection. It is recommended the Town of Hadley investigate the proper location for this sign. If in fact the sign has been placed incorrectly, the sign should be removed and installed at the proper location.

Median

The median on the Middle Street southbound approach lacks vertical deflection features or protective measures due to the absence of curbing and the raised pavement surface resulting from repeated repaving. Vegetative growth was observed along the median during the field visit. This median provides refuge for the pole that supports the supplemental midblock STOP sign and Flashing Beacon. It is recommended that this median be repaired, and proper curbing installed along the edges to provide vertical deflection and protection from turning vehicles.

MassDOT informed the PVPC that the flashing beacon along the median was hit but the overhang of the trucks turning left, therefore curbing may not be completely effective in reducing the risk. Additionally, painting lane line along the intersection to guide turning vehicles was recommended as a measure to reduce possibility of such crashes with large trucks with overhang trailers.

Figure 16: Existing Median along Middle Street North of the Intersection



Intersection Control / Warning Flashing Beacon

The pole mounted Intersection Control Beacon or Warning Beacon located along the median at the southbound approach of Middle Street has been damaged by multiple collisions in the past. This beacon or the bollard protecting the beacon is offset from the actual intersection; however, it is still in the path of some of the heavy vehicles which require larger turning radii. It is recommended that in long term the Town of Hadley considers installing this beacon overhead above the intersection along a span wire or a mast arm. This will require an engineering study and removal of the post mounted beacon on the median.

Traffic Calming

Speeding is observed to be one of the issues along Bay Road. Besides increased signage and speed feedback signs, several vertical and horizontal deflection measures are utilized based on suitability to reduce speed along major roads.

Raised Crosswalk

One of the popular vertical separation / improvement measures for slowing down traffic which is promoted by VRU advocacy groups is installing raised or textured crosswalks. These provide improved visibility and awareness to the drivers and have resulted in successfully slowing down traffic along urban roads. If the Town of Hadley would like to pursue this improvement, there are several examples in the region of such crosswalks installed with appropriate pavement markings.

Optical Awareness Measures

Federal Highway Administration's 'Systemic Application of Multiple Low-Cost Countermeasures for Stop-Controlled Intersections'⁶ guidelines outline pavement markings, signing, and visibility / sight distance improvements as the three effective solutions to improve safety. For major street approaches: oversized advance intersection ahead warning signs with retroreflective sheeting on signposts are recommended along with enhanced pavements markings to delineate through lane edge lines. These measures will help reduce the speeds along Bay Road.

⁶ [Systemic Application of Multiple Low-Cost Countermeasures for Stop-Controlled Intersections](#)

Speed Feedback Signs

The Town of Hadley could utilize Speed Feedback Signs in the vicinity of this intersection to alert the drivers about their speed and encourage them to slow down. These signs are placed farther downstream from the study area and have been effective in slowing down traffic to the east of this location.

Vulnerable Road Users Infrastructure

This location has vegetation separated sidewalks and a crosswalk but there are no bicycle lanes or signs in the immediate vicinity. The shoulders along Bay Road are narrow and oftentimes bicycles are observed within the traffic lanes. As shown in the VRU count data, a significant number of VRU traffic was observed during data collection. Therefore, it is advisable to consider several short- and long-term improvement measures to expand VRU infrastructure at this location.

Rectangular Rapid Flashing Beacon (RRFB)

The Town of Hadley has already secured MassDOT's Grant for the installation of an RRFB along the existing crosswalk. This RRFB will include pedestrian head light to further enhance visibility of VRU. This improvement will greatly enhance the visibility of the crosswalk and improve safety and compliance for both motorists and VRUs.

Safe Bicycle Passing Signs or 4-ft VRU Clearance Signs

These signs are a part of the Commonwealth's Vulnerable Road Users (VRU) law, which requires motorists to give a minimum of 4 feet of clearance when passing vulnerable road users. The Town of Hadley has utilized these signs at other locations effectively and it is recommended that they continue to do so especially along Bay Road.

Education and Awareness

MassDOT and Safe Routes to School have summarized a list of best and safe practices⁷ for both motorists and cyclists to be able to share the road. This summary of tips and guidelines is very helpful and comprehensive tool for local communities and advocacy groups to share with schools, libraries, and towns to increase awareness and compliance. It is recommended that the Town of Hadley DPW and the Police Department utilize these handouts during Townwide campaigns and meetings to encourage safe and responsible driving practices in the Town.

⁷ <https://www.mass.gov/doc/srts-bike-friendly-driving-backpack-flyer-english/download>

Shared Use Path or Side Path

Widening the existing sidewalks to install a side path or a shared use path along Middle Street and Bay Road will address the lack of bicycle accommodations. This is a long-term improvement and will require a detailed assessment of removal of trees, right of way clearance, or relocation of utility poles. It is recommended that the Town considers this option as a future course of action when planning for improvements along this location and its intersecting roadways. Constructing a shared used path or other bike facilities should be considered in accordance with the historic preservation of Middle Street's natural features.

The Town of Hadley could evaluate the bus stops along with PVTa in conjunction to widening the sidewalk or adding a side path. This will allow the students to have a proper alighting and departure area.